

Name SolutionsYear 9
Investigation 1Linear Models
In Class Validation

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Baldyvis
Secondary College

You need to organise a plumber after a burst pipe has flooded your bedroom. You have been given three quotes for replacing the broken pipes.

Choice 1 – Peter's Plumbing

Peter charges \$100 per hour and but does not charge an initial start up fee

Choice 2 - Waldo's Water Workers

Waldo charges \$150 flat fee plus \$60 per hour

Choice 3 – Pipes by Bruce

Bruce charges \$100 flat fee plus \$80 per hour

1. Complete the table of values for each option and write a rule using "C" for cost and "N" for Number of hours that could be used for each option.

Peter's Plumbing

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	0	100	200	300	400	500	600

Now write a rule you can use to calculate the cost of Peter's Plumbing.

$$y = 100x$$

Waldo's Water Workers

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	150	210	270	330	390	450	510

Now write a rule you can use to calculate the cost Waldo's Water Workers.

$$y = 60x + 150$$

Pipes by Bruce

N (number of hours)	0	1	2	3	4	5	6
C (cost in \$)	100	180	260	340	420	500	580

Now write a rule you can use to calculate the cost Pipes by Bruce.

$$y = 80x + 100$$

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2. Plot these three options on the graph paper provided and label each option.
3. Does it make sense to draw a line joining the points you have marked on the graph? Give reasons for your answer.

2

Yes, because it is easier to see the pattern and which lines go together.

4. Describe how the three options value for money changes depending on the length of the disco.

2

At first the cheapest is Pete's Plumbing but at around 4 hrs, Waldo's Water Works becomes the cheapest.

5. Which option is the best value if the plumbing work takes 2 hours?

1

Pete's Plumbing

6. Which option is the best value if the plumbing work takes 5 hours?

1

Waldo's Water Works

7. a) An estimate has suggested that the work should take 4 and a half hours. Using your graph, find out what the cost would then be for each of the three options.

3

a) \$450

b) \$420

c) \$400

b) Check that these values are correct by substituting N (number of hours) back into your rule to find the cost.

$$4.5 \times 100 = \$450$$

③

$$60 \times 4.5 + 150 = \$420$$

$$4.5 \times 80 + 100 = \$460$$

8. You are running short of money for the month and only have \$300, For how much time can you afford to pay Waldo's Water Workers?

$$2 \text{ hours} \checkmark = \$270 \text{ with } \$30 \text{ spare} \checkmark$$

9. Your flat mate has found an additional \$100. How many hours can you now afford to pay Waldo's Water Workers?

$$4 \text{ hrs} \checkmark = \$390 \text{ with } \$10 \text{ spare} \checkmark$$

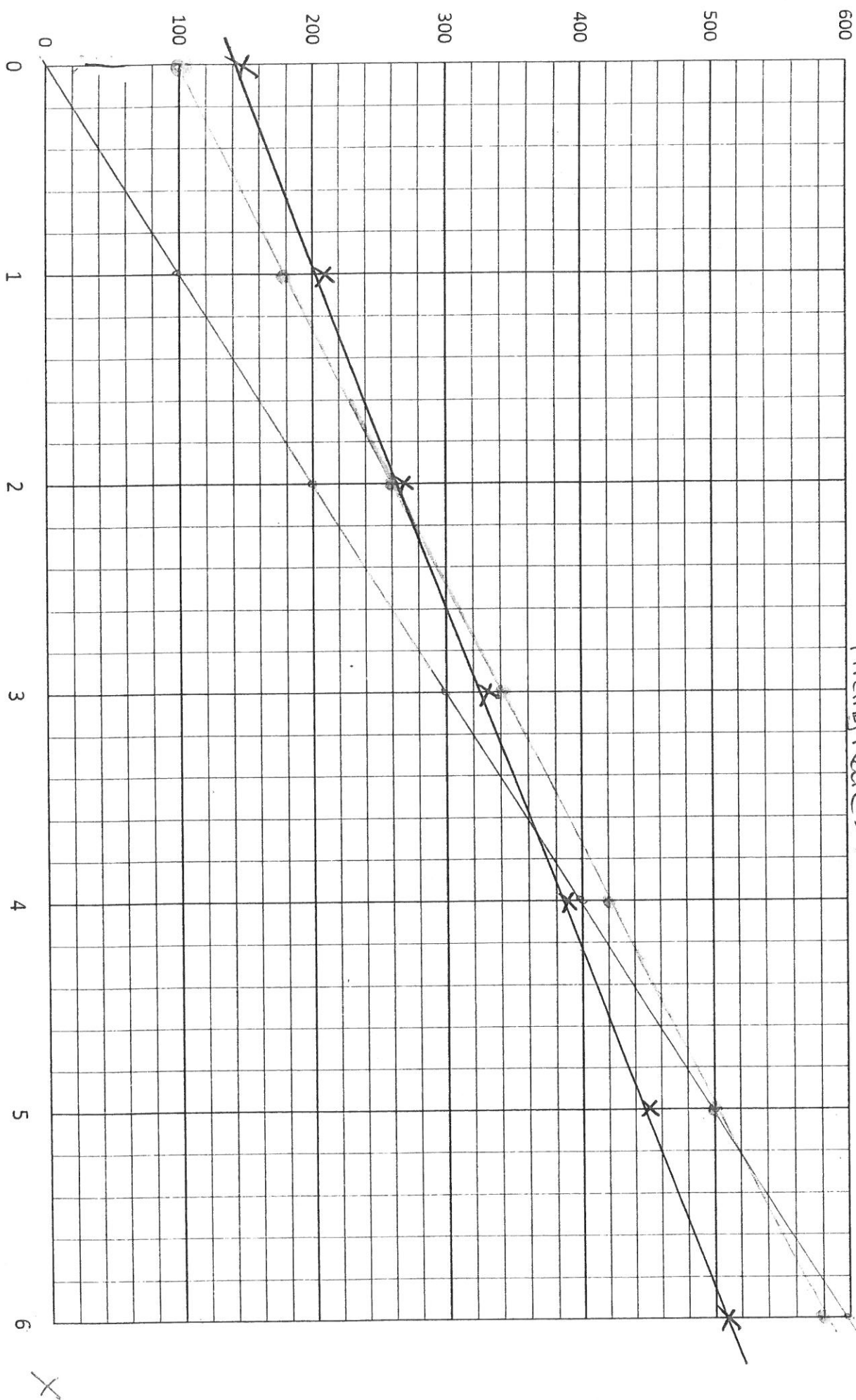
10. Given the amount of money you have (including the extra \$100), is there a better option than Waldo?

No. There is not a better option as you save \$10 with Waldo's.

Pete's same hrs costs \$10 more +

Pipes by Bruce gets you ^{only} 3 hrs worth.

(dollars \$) Cost



(Plumber's Comparison Pricing Rate)

Pete's Plumbing
Waldos Water Works.

Time (hours)